Field goals and logistic regression (answer key)

Does the chance of kicking a paper field goal depend on distance? Each of you kicked once from a random distance between 10 and 50 cm and recorded whether you made it. Now we'll model the class's data.

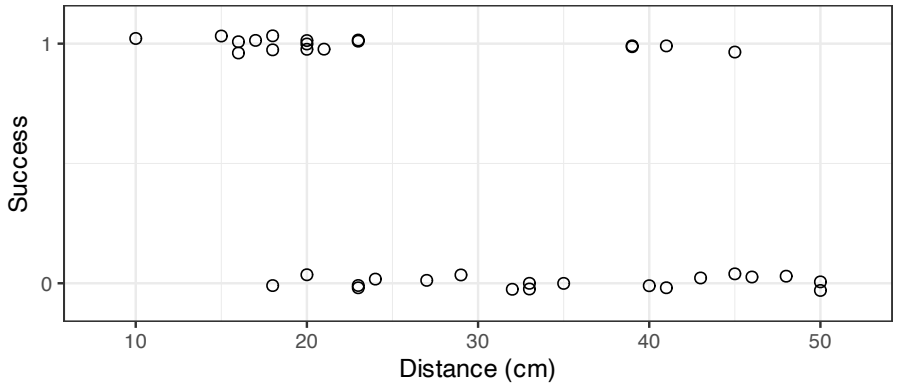
Placeholder: this key uses simulated data. Save the class's responses as data/field-goals.csv and re-render.

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \qquad p = \frac{1}{1+e^{-\text{logit}(p)}}$$

- p: probability of success, between 0 and 1
- $\text{logit}(p)$: the log odds of success, any value from $-\infty$ to ∞
- Left: from probability to log odds. Right: from log odds back to probability.

1. Draw the data

Sketch the class's data. Each kick is one point: its distance on the x-axis, and 0 (missed) or 1 (made it) on the y-axis.



2. Write the model

Write the model in statistical notation. What's the distribution of the response? What links p to the linear equation?

$$\text{success} \sim \text{Bernoulli}(p)$$
$$\text{logit}(p) = \beta_0 + \beta_1 \text{dist_cm}$$

3. Write down the coefficients

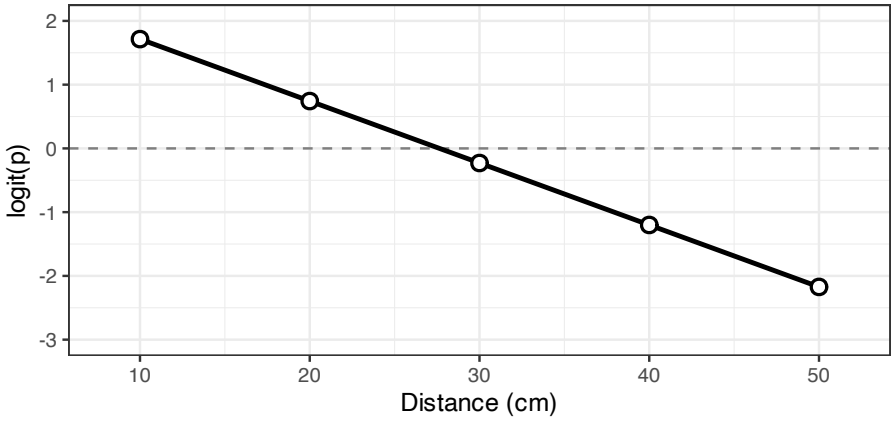
We'll fit the model in R.

```
fg_glm <- glm(success ~ dist_cm, family = binomial(link = "logit"), data = field_goals)
coef(fg_glm)
```

Term	Coefficient	Estimate
`(Intercept)`	β_0	2.686
`dist_cm`	β_1	-0.097

4. Draw the linear part

Draw $\text{logit}(p) = \beta_0 + \beta_1 \text{dist_cm}$. It's a straight line, so you only need two points. Label the y-axis first.



5. Calculate p

For each distance, calculate the log odds from your coefficients. Then convert the log odds to a probability.

Distance (cm)	$\text{logit}(p) = \beta_0 + \beta_1 \times \text{distance}$	$p = \frac{1}{1+e^{-\text{logit}(p)}}$
10	1.71	0.85
20	0.74	0.68
30	-0.23	0.44
40	-1.20	0.23
50	-2.17	0.10

How much does $\text{logit}(p)$ change between each row? -0.97

How much does p change between each row? -0.17, -0.23, -0.21, -0.13

6. Draw the probabilities

Plot your five values of p, then connect them with a smooth curve. Add the data from Part 1.

